

Optimal Intraday Delta Hedging

Static Band Baseline with TWAP/VWAP Execution

Problem Setup

The Liq+ book accumulates net delta exposure $\Delta(t)$ continuously throughout the trading day from JPM's options executions. The desk must be **delta-flat overnight**: $\Delta(T) = 0$. The central trade-off is:

Action	Benefit	Cost
Hedge immediately	Minimal delta risk	High transaction and market impact cost
Do not hedge	Minimal cost	Large residual delta risk
Hedge systematically	Balanced	Requires clear trigger and execution rule

The original framework considered a dynamic Whalley-Wilmott band, an Almgren-Chriss execution engine, and a possible stochastic arrival model for future delta flows. These models are useful as long-run extensions, but the first implementation should be simple, empirical, and directly backtestable with available market data.

Therefore, the first-stage framework uses:

- a static no-trade band around a zero target delta;
- TWAP and VWAP as initial execution baselines;
- Bloomberg-available data to estimate execution cost and delta risk;
- an explicit objective function to choose the best band width and execution method.

Practical goal: The first stage is not to claim a fully optimal theoretical strategy. The goal is to build a clean empirical baseline that can be implemented and tested. Once this baseline is working, we can further optimize the band and execution schedule.

Step 1 — Static Band Filter

Overview

At every point in time, the system monitors the real-time net delta $\Delta(t)$ of the entire options book. Instead of using a dynamic target in the first version, we set the target delta to a static value:

$$\Delta_{\text{target}} = 0. \quad (1)$$

The no-trade region is a static band around this target:

$$-H \leq \Delta(t) \leq H. \quad (2)$$

The system compares the current book delta against this band:

- **Scenario A (Inside band):** $|\Delta(t)| \leq H$. The system does nothing. No spread is crossed and no execution cost is incurred.
- **Scenario B (Band breached):** $|\Delta(t)| > H$. The execution module is triggered. The system computes the hedge order needed to reduce the residual delta exposure.

Trigger Condition

The trigger condition is:

$$|\Delta(t)| > H. \quad (3)$$

Once this condition is satisfied, the hedge order is defined as:

$$Q_t = -\Delta(t). \quad (4)$$

If the book is long delta, the hedge order sells the underlying or hedge instrument. If the book is short delta, the hedge order buys the underlying or hedge instrument.

Band Width as an Empirical Parameter

The band width H is not assumed to be optimal. It is treated as a parameter to be selected through backtesting:

$$H \in \{H_1, H_2, \dots, H_m\}. \quad (5)$$

For example, the grid can be defined in delta shares or delta notional:

$$H \in \{25,000, 50,000, 100,000, 200,000\}. \quad (6)$$

A narrow band triggers more frequent hedging. This lowers delta risk but increases execution cost. A wide band triggers less frequent hedging. This saves execution cost but increases residual delta risk.

Why static first? A static band is easy to explain, implement, and backtest. It also creates a clean baseline. Dynamic Whalley-Wilmott bands can be introduced later after the empirical cost-risk baseline is established.

Step 2 — TWAP/VWAP Execution Engine

Overview

Once the static band is breached, the strategy must decide how to execute the hedge order Q_t . In the first-stage implementation, we do not use the full Almgren-Chriss framework. Instead, we compare two simple and observable execution baselines:

- **TWAP:** split the order evenly over time.
- **VWAP:** split the order according to expected market volume.

This simplification makes the first backtest more practical. TWAP and VWAP are standard execution benchmarks and can be implemented using Bloomberg price and volume data.

TWAP Execution

Suppose the hedge order Q_t is executed over N intervals. Under TWAP, the same quantity is traded in each interval:

$$q_k^{\text{TWAP}} = \frac{Q_t}{N}, \quad k = 1, 2, \dots, N. \quad (7)$$

TWAP is robust and easy to implement because it does not require a volume forecast. Its main weakness is that it does not adjust to the intraday liquidity profile.

VWAP Execution

Under VWAP, the strategy trades more during periods with higher expected market volume. Let $\hat{w}_k$ be the expected fraction of total volume in interval k , with:

$$\sum_{k=1}^N \hat{w}_k = 1. \quad (8)$$

The VWAP child order is:

$$q_k^{\text{VWAP}} = Q_t \hat{w}_k, \quad k = 1, 2, \dots, N. \quad (9)$$

The weights $\hat{w}_k$ can be estimated from historical intraday volume profiles. VWAP is more liquidity-aware than TWAP because it trades more when the market is naturally more active.

Execution Horizon

Let τ denote the execution horizon after the trigger. The initial backtest can test a small grid of horizons:

$$\tau \in \{15 \text{ minutes}, 30 \text{ minutes}, 60 \text{ minutes}\}. \quad (10)$$

A shorter horizon reduces delta risk but may increase execution cost. A longer horizon may reduce execution cost but leaves delta exposure open for more time.

Book Delta Update

The book delta changes due to new option executions and hedge trades. We update the net delta using the accounting identity:

$$\Delta(t) = \Delta(t^-) + \Delta_{\text{new}}(t) + \Delta_{\text{hedge}}(t). \quad (11)$$

where:

- $\Delta(t^-)$ is the book delta immediately before the new event;
- $\Delta_{\text{new}}(t)$ is the delta added by new option executions;
- $\Delta_{\text{hedge}}(t)$ is the delta change from hedge trades.

After each update, the strategy checks the trigger condition in equation (3). If the band is breached while an existing TWAP or VWAP schedule is still active, the baseline rule is to cancel the remaining schedule and replace it with a new schedule based on the updated residual delta.

Practical rule: The execution schedule is always based on the current net delta of the book. This avoids carrying an outdated execution schedule when new delta arrives.

Objective Function

The objective is to choose the hedging rule that best balances execution cost and delta risk. For a given band width H , execution method e , and execution horizon τ , define the total loss as:

$$L(H, e, \tau) = C_{\text{exec}}(H, e, \tau) + \lambda_{\text{risk}} C_{\text{risk}}(H, e, \tau). \quad (12)$$

where:

- H is the static no-trade band width;
- $e \in \{\text{TWAP}, \text{VWAP}\}$ is the execution method;
- τ is the execution horizon;
- C_{exec} is the total execution cost;
- C_{risk} is the total intraday delta risk;
- λ_{risk} is the risk penalty parameter.

The empirical optimization problem is:

$$(H^*, e^*, \tau^*) = \arg \min_{H, e, \tau} [C_{\text{exec}}(H, e, \tau) + \lambda_{\text{risk}} C_{\text{risk}}(H, e, \tau)]. \quad (13)$$

This formulation makes the project directly testable. Instead of assuming an optimal band or schedule, we use historical backtests to choose the combination that minimizes the combined cost-risk loss.

Execution Cost Measurement

Bloomberg-Based Proxy

Because actual execution data is not available, we estimate execution cost using a Bloomberg-based proxy. The execution cost has two components:

$$C_{\text{exec}} = C_{\text{spread}} + C_{\text{impact}}. \quad (14)$$

The first component is the cost of crossing the bid-ask spread. For each child order q_k , spread cost is approximated by:

$$C_{\text{spread}} = \sum_k |q_k| \cdot \frac{\text{Ask}_k - \text{Bid}_k}{2}. \quad (15)$$

ADV-Adjusted Market Impact

The second component is a market impact proxy. Mentor feedback emphasized that transaction cost should depend on the trade size relative to market liquidity, not only on the absolute number of shares traded. Therefore, we use average daily volume (ADV) to scale impact.

The participation ratio is:

$$\rho_k = \frac{|q_k|}{ADV_k}. \quad (16)$$

A simple ADV-adjusted impact proxy is:

$$C_{\text{impact}} = \sum_k \alpha S_k |q_k| \left(\frac{|q_k|}{ADV_k} \right)^\beta. \quad (17)$$

where S_k is the underlying price, ADV_k is average daily volume, and α, β are sensitivity parameters.

Therefore, the total execution cost proxy is:

$$C_{\text{exec}} = \sum_k |q_k| \left[\frac{\text{Ask}_k - \text{Bid}_k}{2} + \alpha S_k \left(\frac{|q_k|}{ADV_k} \right)^\beta \right]. \quad (18)$$

(If Bloomberg TCA is available, it can be used either as a direct execution cost input or as a validation check for this proxy.)

Interpretation: Trading 100,000 shares is not equally costly across underlyings. If ADV is high, the participation rate is small and expected impact should be low. If ADV is low, the same order represents a larger share of market volume and should receive a higher impact penalty.

Delta Risk Measurement

Delta risk measures the cost of leaving residual delta unhedged during the day. We use volatility-scaled squared delta exposure:

$$C_{\text{risk}} = \sum_{j=1}^M \Delta(t_j)^2 \sigma(t_j)^2 \Delta t. \quad (19)$$

where $\Delta(t_j)$ is the residual book delta at time t_j , $\sigma(t_j)$ is the volatility input, and Δt is the length of each time interval.

Symbol Summary

Symbol	Description	Source
H	Static no-trade band width	Chosen from backtest grid
e	Execution method, $e \in \{\text{TWAP}, \text{VWAP}\}$	Strategy choice
τ	Execution horizon	Chosen from backtest grid
Q_t	Parent hedge order generated after band breach	$Q_t = -\Delta(t)$
q_k	Child order size executed in interval k	TWAP/VWAP schedule
S_k	Underlying price in interval k	Bloomberg
Bid_k	Bid price in interval k	Bloomberg
Ask_k	Ask price in interval k	Bloomberg
Mid_k	Mid price in interval k	Bloomberg
V_k	Intraday market volume in interval k	Bloomberg
$\hat{w}_k$	VWAP volume weight in interval k	Estimated from V_k
ADV_k	Average daily volume	Bloomberg
$\Delta(t_j)$	Residual book delta at time t_j	Book delta update process
$\sigma(t_j)$	Volatility input at time t_j	Bloomberg BVOL or realized volatility
Δt	Length of each time interval	Backtest sampling frequency
α	Market impact level parameter	Sensitivity analysis
β	Market impact curvature parameter	Sensitivity analysis
λ_{risk}	Risk penalty parameter	Tuning parameter
C_{exec}	Total execution cost	Spread cost plus impact proxy
C_{risk}	Total intraday delta risk	Volatility-scaled residual delta exposure
$L(H, e, \tau)$	Combined objective function	Optimization target

The final objective function combines execution cost and delta risk:

$$L(H, e, \tau) = C_{\text{exec}}(H, e, \tau) + \lambda_{\text{risk}} C_{\text{risk}}(H, e, \tau), \quad (20)$$

where

$$C_{\text{exec}}(H, e, \tau) = \sum_k \left[|q_k| \frac{\text{Ask}_k - \text{Bid}_k}{2} + \alpha S_k |q_k| \left(\frac{|q_k|}{ADV_k} \right)^\beta \right], \quad (21)$$

and

$$C_{\text{risk}} = \sum_{j=1}^M \Delta(t_j)^2 \sigma(t_j)^2 \Delta t. \quad (22)$$

Therefore, the empirical optimization problem is:

$$(H^*, e^*, \tau^*) = \arg \min_{H, e, \tau} [C_{\text{exec}}(H, e, \tau) + \lambda_{\text{risk}} C_{\text{risk}}(H, e, \tau)]. \quad (23)$$

Strategies and Benchmarks

The initial backtest separates simple benchmarks from the proposed static-band strategies. The benchmarks provide reference points for the cost-risk trade-off, while the proposed strategies are the rules we aim to evaluate and optimize.

Benchmarks

Benchmark	Description	Purpose
Zero Hedge	Do not hedge intraday; flatten the remaining delta at close.	Lower execution cost bound and upper delta risk bound.
Full Hedge	Hedge every delta arrival immediately.	Lower delta risk bound and upper execution cost bound.

Proposed Static-Band Strategies

Strategy	Description	Purpose
Static Band + TWAP	Hedge only when $ \Delta(t) > H$, then execute the hedge order using TWAP.	Simple implementable baseline with optimized band width and horizon.
Static Band + VWAP	Hedge only when $ \Delta(t) > H$, then execute the hedge order using VWAP.	Liquidity-aware baseline using volume-based execution.

Backtest Procedure

For each trading day, each band width H , execution method e , and execution horizon τ , the backtest follows:

1. Initialize the book delta at the start of the day.
2. Update $\Delta(t)$ when new delta arrivals occur.
3. Check whether $|\Delta(t)| > H$.
4. If the band is not breached, do nothing.
5. If the band is breached, generate hedge order $Q_t = -\Delta(t)$.
6. Execute Q_t using TWAP or VWAP over horizon τ .
7. Update $\Delta(t)$ after each hedge child order.
8. At market close, force any remaining delta to zero.
9. Record execution cost, delta risk, combined loss, number of triggers, total traded volume, and participation ratio.

Optimization and Evaluation

The main output is the combination (H^*, e^*, τ^*) that minimizes the objective function in equation (13). Each strategy is evaluated using:

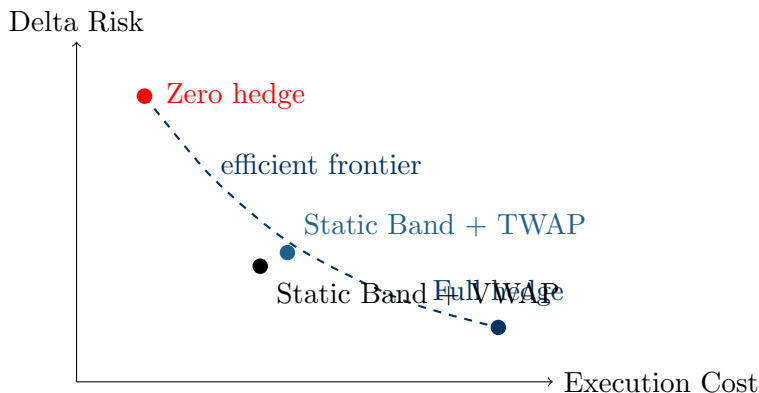
- total execution cost;
- total delta risk;
- combined loss;
- number of hedge triggers;
- total traded volume;
- average participation ratio;
- end-of-day residual delta before forced close.

The key empirical question is:

Can static delta bands combined with TWAP or VWAP execution reduce the combined cost-risk loss relative to full hedge and zero hedge benchmarks?

Efficient Frontier Visualization

The cleanest evaluation is an efficient frontier in (execution cost, delta risk) space:



A better strategy should move the result down and to the left, meaning lower execution cost and lower delta risk.

Future Extensions

After the static band and TWAP/VWAP baseline is implemented, we can optimize the framework further.

Optimized Static Band

The first extension is to select the static band width that minimizes historical combined loss:

$$H^* = \arg \min_H [C_{\text{exec}}(H) + \lambda_{\text{risk}} C_{\text{risk}}(H)]. \quad (24)$$

Dynamic Intraday Band

The static band can later be replaced by a time-dependent band $H(t)$. For example, the band can be wider in the morning and narrower near market close to reflect the increasing importance of the end-of-day delta-flat constraint.

Almgren-Chriss Execution

Once the execution cost proxy is validated, TWAP and VWAP can be compared against a simplified Almgren-Chriss schedule. Instead of fully calibrating all parameters at the beginning, the urgency parameter can be tested over a grid.

Arrival-Aware Execution

If option execution arrival data become available, we can model future delta arrivals and condition execution speed on expected future offsetting flow.

Individual Contributions

Future project updates will include a short individual contribution section to make progress transparent:

Team Member	Contribution
Alex	
Derek	
Timmy Chung	
Esha	
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Summary of Framework

Component	Framework	Answers	Extension
Trigger	Static band around zero delta	When to hedge	Dynamic band $H(t)$
Execution	TWAP / VWAP	How to execute	Almgren-Chriss
Cost	Spread + ADV impact proxy	How to measure trading cost	Bloomberg TCA validation
Risk	Volatility-scaled residual delta	How to measure delta risk	Dynamic risk aversion
Optimization	Minimize execution cost plus risk-weighted delta risk	Which strategy is best	Arrival-aware execution